QA Team Dashboard - Application & Data Flow

System Architecture, Manager & Tester Roles, and Data Flow Specification

1. SYSTEM OVERVIEW & ARCHITECTURE

The QA Team Dashboard is a centralized platform for managing QA testing projects, defect tracking, and daily sprint progress updates under a Single Source of Truth.

- Role-Based Access: Dedicated portals for Managers [/manager/*] and Testers [/tester/*].
- Decoupled Sheets: Google Sheets and Cloud Excel workbooks are stored purely as reference links. Testers click "Live Sheet" to access workbooks without manual count inputs.
- Single Source of Truth: All defect metrics [Total, Open, Retest, Retested, Closed, Crit] are dynamically aggregated directly from the reported bug tickets in the database.

2. MANAGER PERSONA - MODULES & WORKFLOW

A. Manager Overview [/manager/overview]:
Executive cockpit with high-level KPI badges, defect distribution charts by project, recent sprint activity feeds from testers, and urgent blocker notifications.

B. Projects Management [/manager/projects]:
Create new QA projects, assign/unassign QA testers, attach/update live test sheet reference links, and safely delete completed project workspaces.

C. Bugs & Focus Queue [/manager/bugs]:
Live Defect Overview table with quick Live Sheet launchers, and Today’s Active Focus Queue highlighting defects logged today, pending retests, and critical escalations.

D. Master Bug List [/bugs/list]:
Historical defect archive with Category Toggle [Bugs vs. Req Changes], multi-filters [Project, Tester, Severity, Status], full-text search, and ticket deletion.

E. Team Management [/manager/team]:
Add new tester accounts, toggle active/inactive status, and monitor workloads.

3. TESTER PERSONA - MODULES & WORKFLOW

A. Tester Overview [/tester/overview]:
Personal workstation showing assigned projects, personal defect logging metrics [Open, Need Retest, Retested, Closed], and shortcuts to live test sheets.

B. My Assigned Projects [/tester/projects]:
Focused view of projects designated to this tester. Features a "Live Sheet" button to open Google Sheets/Excel in a new tab, and an "Edit Link" button to update URLs.

C. Defect Workbench [/tester/bugs]:
Core test execution page. Testers select an assigned project, log new bugs or requirement changes, advance defect status during retesting, and view sprint charts.
*Critical Bug Escalation: Reporting a critical bug immediately alerts all managers.

D. Daily Progress Submission [/tester/progress]:
Submit daily sprint milestones [Sprint Name, % Progress, Remarks/Blockers].
Instantly notifies the project manager and updates executive overview dashboards.

E. Reported Bug List [/bugs/list]:
Searchable historical audit view across all reported tickets with category toggle.

4. END-TO-END DATA FLOW & LIFECYCLE

Step 1: Project Provisioning -> Manager creates project and assigns tester[s].
Step 2: Test Execution -> Tester opens assigned project, clicks Live Sheet to review test run, and logs bug tickets in the dashboard. If Critical, manager receives alert.
Step 3: Verification Cycle -> Tester transitions ticket: Open -> Need Retest -> Retested -> Closed.
Step 4: Daily Sync -> Tester logs daily % progress; manager dashboard updates in real-time.

5. MATHEMATICAL BALANCE & STATUS MATRIX

Status Lifecycle Definition & Current Verification Formula:

- Open [9]: Defects discovered by QA; awaiting developer triage and code fix.
- Need Retest [2]: Developer deployed fix to test environment; ready for QA validation.
- Retested [5]: QA engineer validated fix in test environment.
- Closed [3]: Verified working and signed-off by QA team.
- Critical [1]: Highest severity blocker tracked as an orthogonal severity dimension.

Mathematical Equality Guarantee:
Total Tracked [19] = Open [9] + Need Retest [2] + Retested [5] + Closed [3]

6. SUMMARY OF TECHNICAL INTEGRATIONS

- Frontend: React 19 SPA with Vite HMR, Glassmorphic CSS Design System.
- Backend: FastAPI REST API with SQLAlchemy ORM and JWT Authentication.
- Query Engine: Single-roundtrip SQL aggregations with indexed B-tree lookups.
- Spreadsheets: Decoupled URL reference pointers with zero manual count entry required.

Generated for QA Team Dashboard Architecture & Flow Review.